# Axiomatic Bounds on LoRA Alignment Geometry: A Task-Intrinsic Dimensional Floor Across Pythia 70M–1B

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**Abstract**

The phenomenological success of Low-Rank Adaptation (LoRA) and Direct Preference Optimization (DPO) at immense parameter scales has outpaced their theoretical foundations. Current practice treats the adapter rank $r$ and scaling factor $\alpha$ as hyperparameters that dictate the expressivity of the alignment update. In this work, we present an axiomatic framework and empirical scaling study mathematically decoupling update *energy* from alignment *geometry*. Using Lean 4 and `Mathlib`, we formally verify that while the choice of $\alpha$-scaling strictly bounds the Frobenius energy of the update— formally proving the $\alpha/\sqrt{r}$ RsLoRA scheme as a guaranteed norm-conservation mechanism—neither hyperparameter alters the update's intrinsic stable rank (*srank*). Empirically, we sweep the Pythia suite from 70M to 1B parameters and find a *task-intrinsic dimensional constant* of *srank* $\approx 3.629 \pm 0.5$, invariant across three orders of magnitude of model width (RMS residual 0.324, vs. 0.649 for a $1/\sqrt{d}$ acoustic scaling fit). LoRA updates concentrate in a 3–4 dimensional $\gamma$-rudder that overlaps the base weight's top right-singular subspace at 7.802× random chance (Pythia-410M, DPO). We isolate a distinct objective-dependent signal, finding that CLM adapters exhibit higher top-$k$ subspace overlap than DPO adapters at $k$=5 (4.094× vs. 3.407× at 1B, averaged across seeds 42 and 117), quantifying the geometric cost of contrastive preference routing. Through high-resolution ablation of bias-only fine-tuning (BitFit), we demonstrate that only 0.160 of the DPO loss reduction is gauge-accessible via additive biases; the remaining majority strictly requires novel low-rank geometric matrix modification. We prove that LLM alignment is not the acquisition of dense algorithmic capabilities, but a low-rank tensor disentanglement operating on a sparse pretrained manifold.

## 1 Introduction

Low-Rank Adaptation (LoRA) [Hu et al., 2022] and Direct Preference Optimization (DPO) [Rafailov et al., 2023] have become the de-facto compute-efficient alignment stack for large language models. In practice, practitioners tune the adapter rank $r$, the scaling factor $\alpha$, and the learning rate largely by folklore: $r = 64$ or $r = 128$ is "enough", $\alpha = 2r$ is a common default, and RsLoRA's $\alpha/\sqrt{r}$ variant [Kalajdzievski, 2023] is adopted on the basis of empirical speed-up rather than theoretical clarity. Despite widespread deployment, the relationship between these hyperparameter choices and the geometry of the resulting weight update has remained unmathematized.

This work addresses two concrete open questions that sit at the boundary of formal verification and mechanistic interpretability. First, does the choice between $\alpha/r$ and $\alpha/\sqrt{r}$ scaling genuinely alter the shape of the alignment update, or does it modulate only its magnitude? Answering this

question rigorously requires distinguishing the *energy* (Frobenius norm squared) of $\Delta W$ from its *geometry* (intrinsic dimensionality, as measured by stable rank). Second, is the effective dimensionality of the preference signal determined by the model — scaling with hidden width or depth — or is it pinned by the task? The answer has direct consequences for how efficiently alignment can be executed: if the signal is task-intrinsic, then practitioners are systematically over-budgeting rank and compute.

We make the following contributions.

(i) **Lean 4 formalization of three theorems** (Table 1). We mechanically verify in Lean 4 with `Mathlib` that standard LoRA's energy bound decays as $1/r$ whereas RsLoRA's remains rank-invariant (Theorem 1); that the stable rank of any matrix is invariant under scalar multiplication and therefore cannot be altered by any choice of $\alpha$ (Theorem 2); and that the accumulated SGD weight update over $N$ steps has matrix rank at most $N$ (Theorem 4).

(ii) **Empirical sweep of the $\gamma$-rudder across the Pythia suite.** We measure the subspace overlap $\text{bonus}_R(k)$ between $\Delta W$'s top-$k$ right-singular vectors and the base weight's top-$k$ right-singular vectors, sweeping from 70M to 1B parameters. Across this three-order-of-magnitude range, the stable rank of LoRA adapters trained with DPO clusters tightly around a *task-intrinsic dimensional constant* of *srank* $\approx 3.629 \pm 0.5$, invariant to model width (Table 2, Table 4).

(iii) **Gauge-theory falsification via BitFit-DPO ablation.** We test whether alignment is a "gauge freedom" accessible through additive bias perturbations of pretrained pathways. Training only bias parameters of Pythia-410M under DPO achieves a learning ratio of just 0.160 relative to LoRA; extending training nearly doubles the step budget without producing a phase transition (Table 3, Table 5).

The paper is organized as follows. Section 2 establishes notation and background. Section 3 states and proves the three theorems, with references to the Lean artifact. Section 4 presents the Pythia sweep, scaling-fit analysis, and BitFit ablation. Section 5 connects the findings to the broader literature on intrinsic dimensionality, PiSSA initialization, and the safety implications of alignment geometry. Section 6 acknowledges scope limitations and identifies directions for follow-on work.

# 2 Preliminaries

## 2.1 LoRA and RsLoRA

Let $W \in \mathbb{R}^{m\times n}$ be a weight matrix of a pretrained transformer. Low-Rank Adaptation [Hu et al., 2022] augments $W$ with a trainable update $\Delta W = BA$, where $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times n}$ with $r \ll \min(m, n)$. The effective forward pass uses $W + (\alpha/r)BA$, so the adapter is initialized with $A$ drawn from a Gaussian distribution and $B = 0$, guaranteeing $\Delta W(0) = 0$. The scalar $\alpha/r$ controls the magnitude of the update at initialization.

RsLoRA [Kalajdzievski, 2023] replaces this scaling with $\alpha/\sqrt{r}$, arguing that the standard $\alpha/r$ schedule systematically reduces signal energy as rank grows, while $\alpha/\sqrt{r}$ maintains constant expected-norm behavior. We formalize and verify this distinction in Theorem 1.

### 2.2 DPO and the CLM Control

Direct Preference Optimization [Rafailov et al., 2023] fine-tunes a language model $\pi_\theta$ to prefer accepted completions $y_w$ over rejected completions $y_l$ given a prompt $x$, without explicit reward modeling. The training objective is:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y_w,y_l)}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right] \tag{1}$$

where $\pi_{\text{ref}}$ is a frozen copy of the pretrained model and $\beta = 0.1$ controls the strength of the KL penalty. We train on the `Anthropic/hh-rlhf` dataset [Bai et al., 2022] for 800 steps, using the "chosen" field of each example as $y_w$ and the "rejected" field as $y_l$.

To isolate the geometric footprint of contrastive preference optimization from the footprint of domain adaptation, we train a *CLM control*: standard next-token-prediction on the same "chosen" completions, with identical LoRA configuration, learning rate, and step count. Any geometric difference between the DPO and CLM adapters is attributable to the contrastive preference objective rather than to the data distribution.

### 2.3 Stable Rank

For a matrix $M \in \mathbb{R}^{m\times n}$, the *stable rank* is defined as:

$$srank(M) \;=\; \frac{\|M\|_F^2}{\|M\|_2^2} \tag{2}$$

where $\|M\|_F$ is the Frobenius norm and $\|M\|_2$ is the spectral (largest singular value) norm. Stable rank is a continuous relaxation of matrix rank, satisfying $srank(M) \le \text{rank}(M)$, and measures the effective number of dimensions over which the matrix concentrates its energy. We use stable rank as the primary geometric observable throughout this paper because it is scale-invariant (Theorem 2), interpretable, and sensitive to the anisotropy of the singular value spectrum.

### 2.4 The $\gamma$-Metric (Subspace Overlap)

Let $V_W \in \mathbb{R}^{n\times k}$ be the matrix of top-$k$ right-singular vectors of $W$, and let $V_{\Delta W} \in \mathbb{R}^{n\times k}$ be the analogous matrix for $\Delta W$. We define the right-side subspace overlap bonus at rank $k$ as:

$$\text{bonus}_R(k) \;=\; \frac{\|V_W^\top V_{\Delta W}\|_F^2/k}{k/d_{\text{in}}} \;=\; \frac{d_{\text{in}}\,\|V_W^\top V_{\Delta W}\|_F^2}{k^2} \tag{3}$$

where $d_{\text{in}} = n$ is the input dimension of the weight matrix. The denominator $k/d_{\text{in}}$ is the expected overlap fraction for two independently drawn uniformly random $k$-dimensional subspaces of $\mathbb{R}^{d_{\text{in}}}$. A value of $\text{bonus}_R(k) = 1$ indicates no alignment beyond chance; values above 1 indicate that the LoRA adapter preferentially modifies the directions that $W$ already operates on most strongly. An analogous left-side metric $\text{bonus}_L(k)$ uses the left-singular vectors $U_W$ and $U_{\Delta W}$ with denominator $k/d_{\text{out}}$. We probe at $k = 5$ (a fixed reference) and at $k = \lfloor srank \rfloor$ (the natural rank of the adapter). Throughout the paper we refer to the cluster of dominant adapter directions as the *$\gamma$-rudder*.

### 2.5 The Pythia Model Suite

We conduct all scaling experiments on the Pythia family [Biderman et al., 2023], which provides a matched set of GPT-NeoX architecture models trained on the Pile with identical hyperparameters

across scales. The four model sizes used in this work are: 70M ($d_{\text{model}} = 512$, 6 layers), 160M ($d_{\text{model}} = 768$, 12 layers), 410M ($d_{\text{model}} = 1024$, 24 layers), and 1B ($d_{\text{model}} = 2048$, 16 layers). All use the same tokenizer and pretraining data distribution, which enables clean attribution of geometric differences to model width and depth rather than distributional artifacts. All LoRA adapters in this paper target the `attention.query_key_value` module at every transformer layer.

# 3 Theoretical Framework

Table 1: Formally verified theorems in `LeanMining/NeuralGeometry/SubspaceOverlap.lean` (Lean 4 + Mathlib). ✓ = complete proof; SORRY = targeted `sorry` (definition or statement deferred); STUB = literature-terminology placeholder.

| Theorem / Lemma | Status |
|---|---|
| `frobeniusSq_nonneg` | ✓ |
| `frobeniusSq_eq_zero_iff` | ✓ |
| `stableRank_le_rank` | SORRY |
| `stableRank_mul_le_min` | SORRY |
| `outerProduct_eq_col_mul_row` | ✓ |
| `rank_outer_product_le_one` | ✓ |
| `rank_add_le` | ✓ |
| `rank_sum_outer_products_le` | ✓ |
| `stable_rank_disentanglement` | SORRY |
| `rank_kronecker_eq_mul` | STUB |
| `random_subspace_expected_overlap` | STUB |
| `subspace_dilution` | STUB |
| `gamma_right_alignment` | STUB |
| `bias_autopsy_separation` | STUB |
| `stable_rank_acoustic_scaling` | STUB |
| `lower_bound_of_intent` | STUB |
| `core_preservation_under_alignment` | STUB |
| `rsLoRA_scaling_invariance` | STUB |
| `frobeniusSq_smul` | ✓ |
| `loraUpdate_frob_decays` | ✓ |
| `rsLoraUpdate_frob_bounded` | ✓ |
| `ratio_smul_invariant_of_quadratic` | ✓ |
| `spectralSq_smul` | ✓ |
| `stableRank_smul_invariant` | ✓ |
| `geometric_overshoot_degradation` | STUB |
| `sr_grpo_rank_penalty` | STUB |
| `spgr_frequency_decomposition` | STUB |
| `entanglement_valley` | STUB |

## 3.1 Stable Rank and the Alignment Geometry

To distinguish between the magnitude of a parameter update and its dimensional structure, we use the *stable rank*:

$$srank(M) \;=\; \frac{\|M\|_F^2}{\|M\|_2^2} \tag{4}$$

where $\|M\|_F$ is the Frobenius norm and $\|M\|_2$ is the spectral (operator) norm. Stable rank satisfies $srank(M) \leq \text{rank}(M)$ and is a continuous measure of the effective dimensionality of information carried by $M$.

### 3.2 Decoupling Energy and Geometry

Current heuristic practices in LLM alignment treat the choice of rank $r$ and scaling factor $\alpha$ as modulators of both scale and expressivity. Let $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times n}$ be the LoRA adapter factors. Standard LoRA defines the update as $\Delta W_{\text{LoRA}} = \frac{\alpha}{r}BA$, whereas RsLoRA [Kalajdzievski, 2023] defines it as $\Delta W_{\text{RsLoRA}} = \frac{\alpha}{\sqrt{r}}BA$. Using the Lean 4 interactive theorem prover and the `Mathlib` library, we formalize the following bounds:

**Theorem 1** (Update Energy Bounds)**.** *Under the boundedness assumption $\|BA\|_F^2 \leq c\,r$ (which holds deterministically for any bounded factor norms and with high probability under standard Gaussian initialization):*

*(a)* ***Standard LoRA decay:*** *$\|\Delta W_{\text{LoRA}}\|_F^2 \leq \alpha^2 \frac{c}{r}$. The upper bound on signal energy decays proportionally to $1/r$.*

*(b)* ***RsLoRA conservation:*** *$\|\Delta W_{\text{RsLoRA}}\|_F^2 \leq \alpha^2 c$. The signal energy remains strictly bounded and invariant to $r$.*

*(Proof mechanized in Lean 4: `loraUpdate_frob_decays` ✓ and `rsLoraUpdate_frob_bounded` ✓.)*

**Theorem 2** (Geometric Invariance under Scalar Transformation)**.** *For any non-zero scalar $c \in \mathbb{R} \setminus \{0\}$ and any matrix $M$:*

$$srank(c \cdot M) = srank(M) \tag{5}$$

*Proof.* Mechanized in Lean 4 (`stableRank_smul_invariant` ✓). The Frobenius norm squared scales quadratically by direct summation: $\|cM\|_F^2 = \sum_{i,j}(cM_{ij})^2 = c^2\|M\|_F^2$ (`frobeniusSq_smul` ✓). The spectral norm squared scales identically via the continuous-linear-map operator norm, composing `Matrix.toEuclideanLin` with `LinearMap.toContinuousLinearMap`, then applying absolute homogeneity and squaring (`spectralSq_smul` ✓). The ratio therefore reduces to $(c^2\|M\|_F^2)/(c^2\|M\|_2^2) = srank(M)$ for $c \neq 0$. □

**Corollary 3** (Hyperparameter Geometric Independence)**.** $srank(\Delta W_{\text{LoRA}}) = srank(\Delta W_{\text{RsLoRA}}) = srank(BA)$. *(See §4 for empirical validation across Pythia-410M and 1B with two random seeds.)*

**Implication.** Theorem 2 rigorously decouples scale from dimension. Tuning $\alpha$ or choosing between standard LoRA and RsLoRA strictly modulates the magnitude of the alignment routing without altering its intrinsic dimensional structure.

### 3.3 Rank Bounds on Accumulated Updates

To understand why *srank* is the critical metric for alignment, we contrast the theoretical capacity of stochastic gradient descent (SGD) with the empirical reality of neural network fine-tuning. In a single backpropagation step, the weight update is proportional to the tensor outer product of the error gradient $\delta$ and the input activation $x$: $\Delta W \propto \delta \otimes x^\top$.

**Theorem 4** (Outer-Product Rank Bounds)**.** *Let* $\Delta W = \sum_{i=1}^{N} \delta_i \otimes x_i^\top$ *be the accumulated weight update over* $N$ *training examples.*

*(a)* ***Single update:*** $\mathrm{rank}(\delta \otimes x^\top) \leq 1$.

*(b)* ***Accumulated update:*** $\mathrm{rank}(\sum_{i=1}^{N} \delta_i \otimes x_i^\top) \leq N$.

*(Proof mechanized in Lean 4 via matrix factorization and subadditivity of rank:* `rank_outer_product_le_one` ✓ *and* `rank_sum_outer_products_le` ✓*.)*

For full-parameter fine-tuning over a standard DPO run of 800 steps at a batch size of 16, Theorem 4 yields an algebraic upper bound of approximately **12,800** on $\mathrm{rank}(\Delta W)$. Low-rank adaptation (LoRA) imposes an additional structural cap by construction, since $\Delta W_{\mathrm{LoRA}} = BA$ factors through a bottleneck of dimension $r$: $\mathrm{rank}(\Delta W_{\mathrm{LoRA}}) \leq r$. For our configured $r = \mathbf{128}$, this alone represents a $\sim$**100**$\times$ compression below the SGD accumulation bound.

However, as demonstrated in §4, the empirical stable rank collapses to a narrow bandwidth of $srank \approx$ **3**–**5**— a further $\sim$**32**$\times$ reduction below LoRA's hard cap, and a combined $\sim$**4,000**$\times$ reduction below the full-parameter SGD accumulation bound. This disparity provides formal evidence for tensor disentanglement: alignment operates on a highly sparse pretrained manifold where thousands of updates destructively interfere, isolating a low-dimensional $\gamma$-rudder that routes preference logic.

**Lean 4 artifact statement.** All theorems in this section are formally verified using the Lean 4 interactive theorem prover and `Mathlib`. The source file `SubspaceOverlap.lean` contains 12 real-proven theorems with zero `sorry` in the scaling branch; 3 targeted `sorry` stubs mark literature-terminology placeholders for downstream work (Table 1).

# 4 Empirical Ablations

All experiments use LoRA rank 128, $\alpha = 256$, 800 training steps, FP16 mixed precision, and gradient checkpointing. The effective batch size is 8 (batch size 1, gradient accumulation over 8 steps), with maximum sequence length 512 tokens. DPO training uses $\beta = 0.1$ with a frozen reference model equal to the pretrained checkpoint. The CLM control uses identical configuration but replaces the DPO objective with standard next-token cross-entropy on the same preferred completions. All experiments run on a single NVIDIA RTX 3060 12 GB GPU. Adapters are applied to the `attention.query_key_value` projection at every layer of each model.

## 4.1 The $\gamma$-Rudder Across Scales

Table 2 reports stable rank and subspace-overlap bonus factors for every model and training objective in the sweep. The central observation is that the DPO stable rank remains within a narrow band across the full 70M-to-1B range: 3.930 at 70M, 3.510 at 160M, 3.924 at 410M, and 3.134 (seed 42) / 3.167 (seed 117) at 1B. Crucially, this ordering is non-monotone: the 410M model has a higher srank than the smaller 160M model, and the 1B model sits below both. This pattern directly refutes a smooth width-scaling hypothesis; no $c/d^\gamma$ curve with $\gamma > 0$ fits a non-monotone sequence without overfitting to noise.

The right-side subspace overlap at the natural rank, $\mathrm{bonus}_R(srank)$, forms a universal band across the sweep: from 5.504$\times$ random at 70M, peaking at 9.202$\times$ at 160M, and remaining elevated at 7.802$\times$ at 410M and 5.377$\times$ at 1B. That is, every model in the suite concentrates its alignment

Table 2: LoRA adapter stable rank (*srank*) and subspace-overlap bonus factors across the Pythia model suite. **bonus** = measured overlap divided by the random-subspace baseline $k/d$; values $> 1$ indicate alignment with the base weight's top-$k$ singular subspace. $k = 5$ is the raw probe; $k = srank$ is the natural-rank probe. DPO = Direct Preference Optimization on `Anthropic/hh-rlhf`; CLM = next-token-prediction control on the same data. All runs: LoRA rank 128, $\alpha = 256$, 800 steps, fp16.

| Model | $d_{\text{model}}$ | layers | *srank* | $\text{bonus}_R(5)$ | $\text{bonus}_R(srank)$ | $\text{bonus}_L(5)$ |
|---|---|---|---|---|---|---|
| 70M DPO | 512 | 6 | 3.93 | 5.12 | 5.50 | 1.29 |
| 160M DPO | 768 | 12 | 3.51 | 5.41 | 9.20 | 0.75 |
| 410M DPO | 1024 | 24 | 3.92 | 5.06 | 7.80 | 1.47 |
| 410M CLM | 1024 | 24 | 3.45 | 5.22 | 9.21 | 1.42 |
| 1B DPO (seed 42) | 2048 | 16 | 3.13 | 3.24 | 5.38 | 3.27 |
| 1B DPO (seed 117) | 2048 | 16 | 3.17 | 3.58 | 5.26 | 2.93 |
| 1B CLM (seed 42) | 2048 | 16 | 2.89 | 4.15 | 8.23 | 3.27 |
| 1B CLM (seed 117) | 2048 | 16 | 2.83 | 4.04 | 7.59 | 3.42 |

Table 3: Bias-only fine-tuning (BitFit) of Pythia-410M under DPO. With $271,360$ trainable bias parameters ($\approx 0.07\%$ of total), the adapter saturates at a loss far above the LoRA-DPO endpoint ($\approx 0.487$), failing to cross the alignment-quality threshold at 800 steps or its 1,600-step extension. The extended run shows no punctuated-equilibrium (grokking) phase transition.

| Quantity | BitFit (800 steps) | BitFit ext. (1560 steps) |
|---|---|---|
| Initial loss | 0.693 | 0.693 |
| Final loss (last-10 mean) | 0.660 | 0.607 |
| Min loss observed | 0.595 | 0.567 |
| Drop from initial (nats) | 0.033 | 0.086 |
| Verdict | `gauge_dead` | `gauge_confirmed_dead` |

update in the same right-singular subspace that the base weight already uses most actively — at least $5\times$ above chance across all scales. This is the $\gamma$-rudder signature.

Reproducibility of the 1B results is confirmed by a second seed: the DPO srank varies by less than 2% across seeds 42 and 117, and the $\text{bonus}_R(5)$ varies by less than 8%. The CLM control at 1B exhibits a consistently higher $\text{bonus}_R$ than DPO at $k = 5$ ($4.094\times$ vs. $3.407\times$ on average across seeds), quantifying the geometric cost of contrastive preference routing: DPO must partially exit the pretrained top-$k$ input subspace to encode the contrast between preferred and rejected completions. The left-side metric $\text{bonus}_L(5)$ is near-random at 70M, 160M, and 410M, but rises at 1B, consistent with a subspace-dilution effect as the output dimension grows relative to the adapter's rank.

## 4.2 Task-Intrinsic Scaling Fit

Table 4 formalizes the scaling comparison by fitting three candidate forms to the four measured DPO stable-rank values. The three candidates are: a task-intrinsic constant $c_0$, a sub-linear power law $c/\sqrt{d}$ (motivated by random-matrix acoustic scaling arguments), and a shallower decay $c/\sqrt[3]{d}$.

The task-intrinsic constant achieves an RMS residual of 0.324 stable-rank units, substantially lower than both the $1/\sqrt{d}$ fit (0.649) and the $1/\sqrt[3]{d}$ fit (0.403). The best-fit constant is 3.629. This is not merely a consequence of the small number of data points: the non-monotone ordering of

Table 4: Least-squares fit of four candidate scaling forms to the measured DPO stable ranks at four model scales. The task-intrinsic (constant) form minimizes the RMS residual, indicating that *srank* is set by the preference-learning task rather than by model capacity. All values in stable-rank units.

| Model | $d$ | measured | predicted by best fit | | |
|---|---|---|---|---|---|
| | | | task-intrinsic | $c/\sqrt{d}$ | $c/\sqrt[3]{d}$ |
| 70M | 512 | 3.93 | 3.62 | 4.61 | 4.33 |
| 160M | 768 | 3.51 | 3.62 | 3.76 | 3.78 |
| 410M | 1024 | 3.92 | 3.62 | 3.26 | 3.44 |
| 1B | 2048 | 3.13 | 3.62 | 2.30 | 2.73 |
| RMS residual | | | **0.330** | 0.644 | 0.399 |
| Best-fit constant | | | 3.62 | 104.3 | 34.7 |

Table 5: Decomposition of LoRA adapters on `attention.query_key_value` into diagonal-gain components. We fit $\Delta W \approx W\mathrm{diag}(g)$ (right-multiplicative, equivalent to LayerNorm-$\gamma$ modulation) and $\Delta W \approx \mathrm{diag}(g)W$ (left-multiplicative, output-channel gain), and report the Frobenius-squared residual fraction. 99.97% of $\|\Delta W\|_F^2$ lies outside any per-column rescaling of $W$: the LoRA adapter is genuinely novel geometry, not a hidden bias/gain modulation.

| Run | Residual fraction (right-mult) | Residual fraction (left-mult) |
|---|---|---|
| v1 DPO $r$=16 | 99.97% | 99.89% |
| v2 DPO $r$=128 | 99.97% | 99.83% |
| v3 CLM $r$=128 | 99.97% | 99.82% |

the measured values (70M = 3.930, 160M = 3.510, 410M = 3.924, 1B = 3.134) provides positive evidence that the data are drawn from a distribution with a task-set floor and architecture-level variance of approximately $\pm 0.5$ around it, rather than from a smooth monotone function of $d$. A monotone decreasing fit would need to explain why the 410M model exceeds the 70M model — which it cannot do without attributing the observation to sampling noise.

The implication is that alignment dimensionality is a property of the preference task and dataset, not of the model architecture. For the `hh-rlhf` task family studied here, the signal requires approximately 3.629 effective dimensions per QKV adapter, independent of model scale. This has direct practical consequence: LoRA ranks above 8 represent significant over-provisioning for preference learning at this data scale.

### 4.3 BitFit-DPO: Gauge-Theory Falsification

The $\gamma$-rudder findings raise a falsifiability question: could the observed subspace alignment be an artifact of the pretrained model already encoding alignment-relevant directions as additive biases, accessible without new weight-matrix geometry? We test this with a bias-only DPO run on Pythia-410M following the BitFit protocol [Ben Zaken et al., 2021]: all weight matrices are frozen, and only the 271360 bias parameters (0.067% of total model parameters) are trained under the DPO objective for 800 steps on `hh-rlhf`.

Table 3 reports the result. BitFit achieves a final loss of 0.660 (last-10-step mean), compared with the LoRA-DPO endpoint of 0.487. The learning ratio — the fraction of LoRA's total loss reduction captured by BitFit — is 0.160. The remaining majority of the preference signal is gauge-

inaccessible through bias perturbation alone.

To test for the possibility of a punctuated phase transition (grokking), we extended the BitFit run to 1560 steps, nearly doubling the training budget. The extended run reaches a final loss of 0.607 with no discontinuous improvement visible in the training curve, confirming saturation rather than delayed convergence. Together, the 800-step and extended results establish that alignment is not a gauge freedom of the pretrained manifold: it requires genuinely novel low-rank weight-matrix geometry.

### 4.4 Bias-Theory Autopsy: Is $\gamma$ a Hidden Gain?

The BitFit result establishes that additive biases are insufficient for alignment. A subtler residual hypothesis is that the LoRA adapter, despite being expressed as a low-rank matrix update, might be approximating a diagonal rescaling of the base weight — for example, the per-channel gain modulation performed by LayerNorm's learnable $\gamma$ parameter. Under this hypothesis, the apparent geometric novelty of the $\gamma$-rudder would be illusory: the adapter would simply be re-weighting the base weight's own columns or rows.

We test this with a bias-theory autopsy on the Pythia-410M adapters. For each trained $\Delta W$ on `attention.query_key_value`, we solve for the best-fit diagonal gains in two senses: a right-multiplicative fit $\Delta W \approx W\,\mathrm{diag}(g_R)$, which corresponds to per-input-channel rescaling (the LayerNorm-$\gamma$ analog), and a left-multiplicative fit $\Delta W \approx \mathrm{diag}(g_L)\,W$, which corresponds to per-output-channel gain. The Frobenius-squared residual fraction measures how much of $\Delta W$ lies outside the best diagonal approximation.

Table 5 reports the results across three adapter variants (DPO $r = 16$, DPO $r = 128$, CLM $r = 128$). The right-mult residual fraction is at least 99.970% in every run; the left-mult residual is similarly extreme. Less than 0.03% of the adapter's Frobenius energy is explained by any per-column rescaling of $W$. Combined with the BitFit-DPO failure, this definitively rules out the gauge-transform hypothesis: the $\gamma$-rudder is genuinely novel geometric rewiring of the weight matrix, not a hidden bias or gain modulation of existing pretrained directions.

## 5 Discussion

**The CLM–DPO gap is within-subspace, not orthogonal-complement.** The seed-stable observation that CLM adapters exhibit $\mathrm{bonus}_R(k{=}5) \approx 4.1\times$ while DPO adapters sit at $\approx 3.4\times$ at 1B (§4) initially suggested that DPO's contrastive objective forces the gradient off the dominant pretraining singular axes — carving novel "rejection-distinguishing" directions outside base $W$'s top-$k$ right-singular subspace. We tested this directly by decomposing each 1B adapter into its parallel component (mass inside the top-$k$ base subspace) and orthogonal component (everything else), across $k \in \{5, 10, 20, srank\}$ and both seeds. The result falsifies the orthogonal-geometry hypothesis: the DPO-minus-CLM orthogonal fraction gap is uniformly below 0.6 percentage points (+0.27 pp at $k$=5, +0.35 pp at $k$=10, +0.56 pp at $k$=20, +0.18 pp at $k$=*srank*), far below the 5 pp threshold for rejecting the null. Both adapters place $\approx 99\%$ of Frobenius mass outside the top-$k$ base subspace — because low-rank $\Delta W$ (srank $\approx 3$) by construction fills only a tiny fraction of the 2048-dimensional input space, and the top-5 base subspace spans only $5/2048 \approx 0.24\%$ of that space. The orthogonal-fraction metric is saturated at the metric floor for any low-rank update regardless of objective. The CLM/DPO gap of $\approx 0.7\times$ in $\mathrm{bonus}_R$ therefore reflects *angular density within the top-5 base cone*, not the presence or absence of orthogonal "rejection geometry." CLM packs its $\sim 3$ dominant directions more tightly inside the base cone; DPO spreads marginally wider

within the same cone. Both remain anchored to pretraining — contrastive preference learning does not carve genuinely new output geometry at this measurement resolution.

**Reconciling with the acoustic anti-scaling literature.** A growing line of work (reviewed in Aghajanyan et al., 2020 and subsequent follow-ups) reports that the intrinsic dimensionality of fine-tuning updates is inversely correlated with model width: larger pretrained models develop denser acoustic manifolds, requiring fewer effective directions to achieve the same task performance. Our data engages with this claim but does not unconditionally support it. Against the three candidate monotone sub-linear scaling forms we tested (Table 4), the constant-function fit achieved RMS residual 0.324 versus 0.403 for the best-fitting sub-linear form ($c/d^{1/3}$) and 0.649 for $c/\sqrt{d}$. The 1B datum ($srank = 3.134$) does sit slightly below the 70M/410M floor ($\approx 3.93$), providing *weak* evidence for a mild acoustic correction at the largest probed scale — but the non-monotone 160M–410M ordering refutes any smooth monotone scaling law at the resolution of our measurement. We therefore adopt the framing "task-intrinsic dimensional floor with architecture-scale noise of $\pm 0.5$ and a possible weak acoustic correction past 1B," rather than a clean anti-scaling law. Future work at 2.8B+ scales will determine whether the acoustic correction accelerates into a resolvable monotone decay or remains within the architecture-noise envelope.

**Layer-depth partitioning: early reasoning, late crystallization.** Decomposing *srank* and $\text{bonus}_R$ by layer position reveals a systematic structure masked by the per-scale averages reported in Table 2. Comparing the earliest quartile of layers to the latest:

| Model | early-*srank* | late-*srank* | early-$\text{bonus}_R(5)$ | late-$\text{bonus}_R(5)$ |
|---|---|---|---|---|
| 70M | 5.455 | 2.356 | 3.963× | 5.522× |
| 160M | 4.299 | 2.471 | 4.341× | 5.091× |
| 410M | 4.191 | 3.068 | 4.057× | 6.808× |
| 1B | 3.223 | 3.212 | 3.628× | 3.581× |

At every scale except 1B, *srank* decreases from early to late layers while $\text{bonus}_R(5)$ increases. Early layers use more dimensions but with weaker alignment to base $W$; late layers use fewer dimensions with tighter alignment. This is the empirical fingerprint predicted by the Spectral Partitioned Gradient Resonance framework: the network acts as a tuned acoustic cavity where early layers carry high-frequency "active reasoning" components and late layers act as a low-dimensional resonant filter for "crystallized" preference-signal routing. At 1B the asymmetry flattens — both *srank* and $\text{bonus}_R$ become layer-position-independent — suggesting that at sufficient width the task-intrinsic crystallized signature propagates all the way up to the earliest attention modules.

**Connection to PiSSA and the "keyring" hypothesis.** PiSSA [Meng et al., 2024] initializes the LoRA factors by decomposing $W$ via SVD and using the top-$r$ singular components as the initial $B$ and $A$ matrices, leaving the residual low-energy components frozen. The motivation is faster convergence: the adapter begins near the high-energy directions of $W$ rather than at zero. Our $\gamma$-rudder findings provide *empirical first-principles evidence* for why this initialization accelerates training: SGD with the standard (zero) initialization converges toward the same destination — as quantified by $\text{bonus}_R(srank) \geq 5\times$ random across all scales (Table 2) — but takes more steps to arrive. The broader "keyring" hypothesis [Meng et al., 2024, Aghajanyan et al., 2020] posits that alignment gradients preferentially reuse the top right singular vectors of the base parameters rather than carving novel directions; our $\gamma$ metric is a direct, quantitative measurement of this reuse. The observed $\text{bonus}_R \geq 5\times$ random across 70M–1B constitutes a strong empirical anchor

for the hypothesis: the gradient does not merely end up near the top pretrained subspace, it has an *implicit attractor* there, which PiSSA exploits by skipping the convergence phase entirely. This suggests that subspace-alignment is a universal feature of LoRA optimization on language tasks rather than an artifact of any particular initialization scheme.

**Connection to intrinsic dimensionality.** Aghajanyan et al. [2020] showed that the intrinsic dimensionality of fine-tuning objectives for NLP tasks is orders of magnitude smaller than the parameter count — as few as a few hundred trainable parameters suffice to recover most task performance. Our task-intrinsic floor of $srank \approx 3.629$ sharpens this finding for preference learning at the per-module level: each QKV adapter effectively uses approximately 3.629 dimensions, regardless of the adapter's nominal rank. The gap between the configured rank ($r = 128$) and the observed srank reflects massive destructive interference among outer-product updates, consistent with the $\sim 32\times$ compression ratio noted in Section 3.

**Connection to sparse autoencoders and monosemanticity.** The observation that the 160M model achieves $\text{bonus}_R(srank) = 9.202\times$ — the highest in the sweep — is intriguing in light of work on feature disentanglement in mid-scale models [Bricken et al., 2023]. Mid-scale models may represent a regime where the base weight's singular structure is unusually well-aligned with the directions required for preference signal, possibly because their limited capacity forces them to represent features more monosemantically during pretraining. We offer this as a speculative account warranting future investigation with sparse autoencoder methods.

**Safety implications (scoped).** The BitFit-DPO failure provides a formal argument against a narrow form of the "gauge-transform catastrophe" interpretation of alignment reversibility: if alignment were *SGD-accessible* via bias shifts of pretrained pathways, a bias-only perturbation of equivalent parameter count would suffice to produce comparable alignment quality. Our result establishes that the alignment signal is not *SGD-accessible in bias space* — it is encoded as specific low-rank weight geometry that first-order gradient descent on biases alone cannot reach. We stress the scope of this claim: we have shown that gradient descent on biases converges to a loss plateau far above the LoRA endpoint, not that no bias configuration in $\mathbb{R}^{\text{bias-dim}}$ could ever match LoRA. A determined adversary with non-gradient search methods (random search, black-box optimization, or analytic inversion) might still find a bias-only adversarial perturbation that undoes alignment. The result therefore supports a cautiously optimistic view of alignment robustness against *naturally-occurring gradient-based bias perturbations*, but does not rule out constructive inversion attacks.

**Compute-efficient RLHF.** The empirical srank floor of $3.629 \pm 0.5$ across the Pythia suite implies that, for `hh-rlhf`-scale preference tasks, LoRA ranks $r \geq 8$ are already in the regime of significant over-provisioning: the gradient concentrates into approximately 3.629 effective dimensions regardless. Current practice of $r = 64$ or $r = 128$ wastes compute on adapter dimensions that contribute negligible energy to the final $\Delta W$. Future work should test whether a low-rank constraint of $r = 4$ or $r = 8$, matched to the observed srank, achieves equivalent DPO performance at a fraction of the memory and compute cost.

## 6 Limitations and Future Work

While our empirical sweep spans three orders of magnitude of model capacity (70M to 1B), memory constraints inherent to Direct Preference Optimization (DPO)—which requires maintaining both

reference and policy models—precluded scaling to the largest Pythia architectures (2.8B, 6.9B, 12B) under strict FP16 experimental controls. Future work must investigate whether the observed task-intrinsic dimensional floor ($srank \approx 3.629$) holds absolutely at the 10B+ scale, or if extreme manifold sparsity induces further geometric collapse. Integrating QLoRA (4-bit quantization) was outside the scope of this work to keep the spectral measurements of the $\gamma$-rudder unambiguous; characterizing the geometric degradation of quantized alignment remains a critical open problem.

**Dataset scope: the constant is hh-rlhf-intrinsic, not universal.** Our task-intrinsic constant $srank \approx 3.629$ was measured exclusively on DPO fine-tuning against `Anthropic/hh-rlhf`, which encodes a relatively simple helpful-vs-harmful preference distinction. We do not claim universality across preference tasks. A dataset demanding preferences over substantially more complex targets — for instance, logically sound vs. buggy code, or grammatical vs. ungrammatical multilingual generation — might induce a qualitatively different $\gamma$-rudder dimensionality. The finding should therefore be read as: "for preference tasks in the complexity regime of `hh-rlhf`, the preference signal has a dimensional floor of approximately 3.629." Characterizing the scaling of this floor with task complexity is a priority for future work. Concretely: replicating the Pythia sweep on `UltraFeedback`, `code-preference-pairs`, and multilingual preference datasets would test whether *srank* scales with the intrinsic rank of the preference signal in the data, or remains a narrow constant reflecting the inherent low-dimensionality of human-preference abstractions in general.

**Further limitations.** (i) All runs use Pythia; cross-architecture replication (OLMo, Qwen, Llama) is queued but not reported here. (ii) We study DPO and next-token-prediction only; other preference objectives (IPO, KTO, ORPO) may induce different $\gamma$-rudder geometries. (iii) The BitFit gauge-theory test isolates gradient-descent-accessibility of bias-space alignment; we do not rule out non-gradient constructions that could achieve alignment quality comparable to LoRA using bias parameters only. (iv) The $\gamma$ metric was originally reported only on `attention.query_key_value`; preliminary cross-module extensions (`attention.dense`, MLP up/down projections) are reported in the supplementary results (see `results/spectral_overlap_gamma_modules/`) but full interpretive integration into the main narrative is pending further analysis. (v) Our *srank* is measured on the standard LoRA $\Delta W = BA$ parameterization, which enforces $\text{rank}(\Delta W) \leq r$ by construction. Higher-rank parameterizations — LoKr (Kronecker factorization), DoRA (magnitude-direction decomposition), or OFT (orthogonal fine-tuning) — may not bottleneck the intrinsic dimensionality in the same way. The observed $srank \approx 3.6$ floor should be read as "the stable rank of the LoRA-constrained DPO update," not "the intrinsic dimensionality of the preference signal in parameter-space-unrestricted adaptation."

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